Mathematical Foundations for Data Science & Computational Intelligence

What the Gradient Doesn't Tell You

Module 5 · Multivariable Calculus · Session 5.3

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Where We Are

Gradient

∇f: one column, one entry per knob. Points steepest uphill.

Jacobian

Many outputs → an m × n matrix. Rows are outputs, columns are inputs.

Still open

≈ 30°

Why did $-\nabla f$ miss the minimum of $x^2 + 3y^2$ at (2, 1)?

Today: finish the chain rule (Thursday's backprop runs on it), then answer the open question.

Quick Check: Answers

$$f(x, y) = x^2y + 3y \quad \text{at } (1, 1): \quad \nabla f = (2, 4)$$

1	Rate of change heading east	**2**
2	Maximum possible rate of change	**$\|(2, 4)\| = \sqrt{20} \approx 4.47$**
3	A direction where f doesn't change	**any direction $\perp (2, 4)$, e.g. $(2, -1)/\sqrt{5}$**

Worked Example

$$\mathrm{f}(\mathrm{x},\,\mathrm{y}) = (\;\mathrm{xy},\;\;\mathrm{x} + \mathrm{y}^2\;)$$

$$J = \begin{bmatrix} \dfrac{\partial f_1}{\partial x} & \dfrac{\partial f_1}{\partial y} \\ \dfrac{\partial f_2}{\partial x} & \dfrac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} y & x \\ 1 & 2y \end{bmatrix} \xrightarrow{(1,\,2)} \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$$

Two Jacobians in Every Network

Linear layer

$$z = Wx \;\Rightarrow\; J = W \qquad W = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} : \quad \begin{aligned} z_1 &= x_1 + 2x_2 \\ z_2 &= x_2 \end{aligned}$$

“The derivative of 3x is 3. The derivative of Wx is W.”

Elementwise activation

$$a = \sigma(z) \;\Rightarrow\; J = \begin{bmatrix} \sigma'(z_1) & 0 & \cdots & 0 \\ 0 & \sigma'(z_2) & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma'(z_n) \end{bmatrix}$$

“Each output only depends on its own input, so everything off the diagonal is 0.”

Shape Cheat Sheet

Function	Derivative	Shape
$\mathbb{R} \to \mathbb{R}$	ordinary derivative	a number
$\mathbb{R}^n \to \mathbb{R}$ (e.g. a loss)	**gradient**	**n-vector**
$\mathbb{R}^n \to \mathbb{R}^m$ (e.g. a layer)	Jacobian	m × n matrix

A gradient is the transpose of a Jacobian with one row (m = 1).

QUESTION FOR THE ROOM

$\mathbf{d/dx\ \sin(x^2 + 1)\ = ?}$

Before differentiating: name the outer and the inner function.

Single-Variable Chain Rule, Slowly

$$g'(x) = \cos(x^2 + 1) \cdot 2x$$

outer derivative, evaluated at the inner × **inner derivative**

Common wrong answer: $\cos(x^2 + 1)$

It forgets the × 2x. In a network, one dropped inner derivative makes the whole gradient wrong.

You Already Used It in Session 5.1

$$L(w, b) = (wx + b - y)^2$$

write $u = wx + b - y$, so $L = u^2$

$$\partial L/\partial w = (dL/du)(\partial u/\partial w) = 2u \cdot x$$
$$\partial L/\partial b = (dL/du)(\partial u/\partial b) = 2u \cdot 1$$

Session 5.1's "error signal × what the knob touched" is the chain rule.
2u is the outer derivative (the error signal). x and 1 are the inner derivatives.

Multivariable Chain Rule

Chain two functions: $\mathrm{x} \;\to\; \mathrm{g} \;\to\; \mathrm{u} = \mathrm{g}(\mathrm{x}) \;\to\; \mathrm{f} \;\to\; \mathrm{z} = \mathrm{f}(\mathrm{u}) = \mathrm{f}(\mathrm{g}(\mathrm{x}))$

$$\underbrace{J_{f\circ g}(x)}_{m\times n} = \underbrace{J_f\big(g(x)\big)}_{m\times k} \cdot \underbrace{J_g(x)}_{k\times n}$$

Chaining functions multiplies their Jacobians.

Shapes: $\mathbb{R}^{\mathrm{n}} \to \mathbb{R}^{\mathrm{k}} \to \mathbb{R}^{\mathrm{m}}$ gives $(\mathrm{m} \times \mathrm{k})(\mathrm{k} \times \mathrm{n}) = \mathrm{m} \times \mathrm{n}$ ✓ The outer Jacobian goes on the left.

Why This Is the Module 2 Payoff

Module 2

Matrix multiplication = composing linear maps

→

Zoom in

Near any point, a smooth function is approximately linear: its Jacobian

→

So

Composing functions ≈ composing linear maps = multiplying Jacobians

The chain rule isn't a new rule: zoom in until everything is linear, then compose.

Worked Example: A Tiny Layer

$W = [1\ 2;\ 0\ 1]$, $x = (1, 1)^T$. Layer: z = Wx, then a = z² (elementwise).

Step	Result
$z = Wx$	$(3, 1)^T$
$a = z^2$	$(9, 1)^T$
$J_a(z) = \mathrm{diag}(2z)$	$\mathrm{diag}(6, 2)$
$J = \mathrm{diag}(6, 2) \cdot W$	$[6\ 12;\ 0\ 2]$

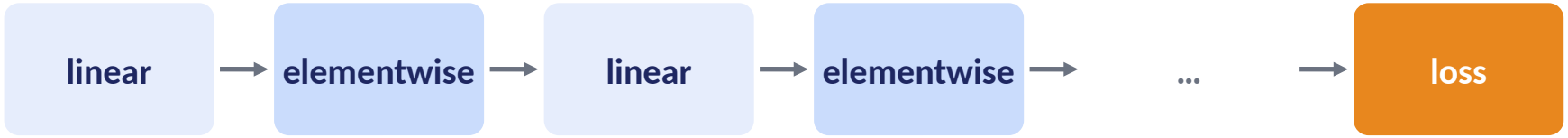
Check one entry directly: $a_1 = (x_1 + 2x_2)^2$
$\partial a_1/\partial x_1 = 6$ ✓ $\partial a_1/\partial x_2 = 12$ ✓

$$J = \underbrace{\begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}}_{J_a(z)=\mathrm{diag}(2z)} \cdot \underbrace{\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}}_{J_z(x)=W} = \begin{bmatrix} 6 & 12 \\ 0 & 2 \end{bmatrix}$$

OPTIONAL · NUDGE CHECK

$x_1 \to 1.01$: $a_1 = 3.01^2 = 9.0601$
$\Delta a_1 / 0.01 \approx 6.01 \approx 6$ ✓

A Deep Network Is This, Repeated



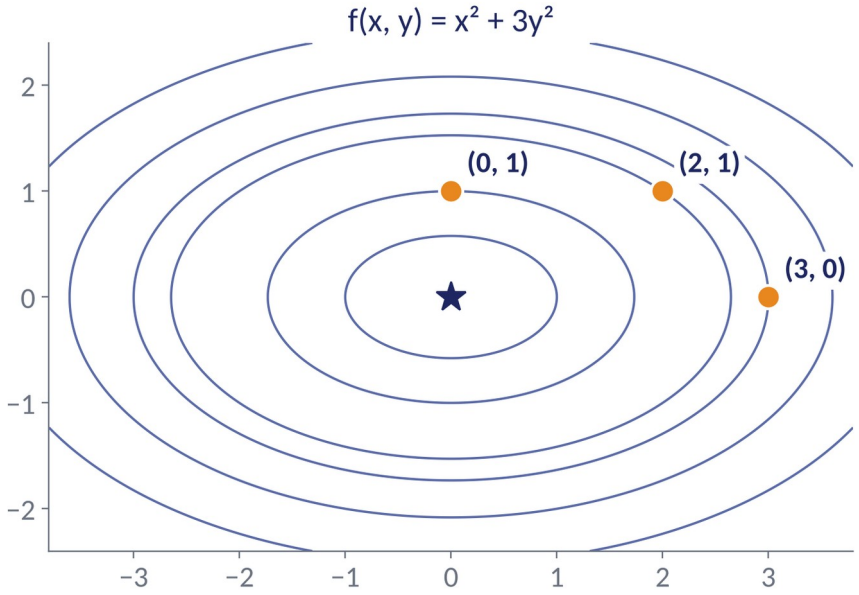
$$J = J_{\text{loss}} \cdot \text{diag}(\sigma') \cdot W_k \cdots \text{diag}(\sigma') \cdot W_2 \cdot \text{diag}(\sigma') \cdot W_1$$

A long product of weight matrices and diagonal matrices.
Each layer adds two factors: its W and a diagonal. Remember this product.

QUESTION FOR THE ROOM

Stand at (0, 1), then at (3, 0). Does $-\nabla f$ point at the minimum?

At (2, 1) it missed by ≈ 30°.

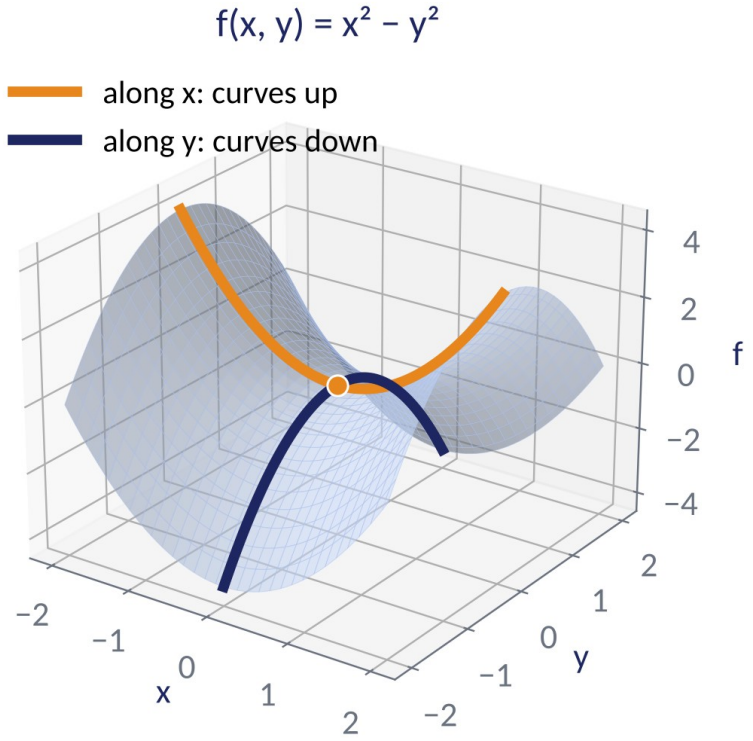


QUESTION FOR THE ROOM

At the bottom of a valley, $\nabla f = 0$. If $\nabla f = 0$, are you at the bottom?

Minimum? Maximum? Neither?

The Saddle



$$\nabla f = (2x, -2y) = 0 \text{ at the origin}$$

A minimum if you walk east–west.

A maximum if you walk north–south.

Gradient zero, and neither. A saddle: like a mountain pass.

So We Need More Information

Minimum

$\nabla f = 0$

valley floor

Maximum

$\nabla f = 0$

hilltop

Saddle

$\nabla f = 0$

mountain pass

The gradient alone can't tell them apart.

In one variable, f'' did (Session 5.1). Now: a second derivative in every direction at once.

The Hessian

$$H = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots \\ \dfrac{\partial^2 f}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \qquad H_{ij} = \frac{\partial^2 f}{\partial x_i \, \partial x_j}$$

Every second derivative, including the mixed ones. n × n.

Symmetric

Mixed partials agree for any well-behaved f.

Jacobian of the gradient

How the slope itself changes as you move.

Three Hessians You Already Know

f(x, y)	H	Shape
$x^2 + 3y^2$	$\mathrm{diag}(2,\ 6)$	the stretched bowl
$x^2 - y^2$	$\mathrm{diag}(2,\ -2)$	the saddle
$-x^2 - y^2$	$\mathrm{diag}(-2,\ -2)$	the hilltop

Valley, saddle, hilltop: the signs on the diagonal already look like the answer. Too easy?

A Hessian That Depends on Where You Are

$$g(x, y) = x^3 - 3x + y^2$$
$$\nabla g = (3x^2 - 3,\ 2y) = 0 \quad \text{at} \quad (1, 0) \quad \text{and} \quad (-1, 0)$$

$$H_g = \begin{bmatrix} 6x & 0 \\ 0 & 2 \end{bmatrix}: \qquad H_g(1,0) = \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}, \qquad H_g(-1,0) = \begin{bmatrix} -6 & 0 \\ 0 & 2 \end{bmatrix}$$

Same function, two critical points, two different Hessians. Which is which? First, the rule.

Practice · Pairs

$$\text{Find the Hessian of } \; f(x, y) = 3x^2 + 2xy + 3y^2$$

Check: is your H symmetric? Where does the 2xy term end up?

2 minutes. We classify it in a few slides.

Same Formula, Upgraded

One variable (Session 5.1)

$$f(x+h) \;\approx\; f(x) + f'(x)\,h + \tfrac{1}{2}\,f''(x)\,h^2$$

Many variables

$$f(\mathbf{x}+\boldsymbol{\Delta}) \;\approx\; f(\mathbf{x}) + \nabla f(\mathbf{x})^{\top}\boldsymbol{\Delta} + \tfrac{1}{2}\,\boldsymbol{\Delta}^{\top} H(\mathbf{x})\,\boldsymbol{\Delta}$$

Live check · $g = x^3 - 3x + y^2$ at (1, 0), step Δ = (0.1, 0.1)
Exact: g(1.1, 0.1) = −1.959 Taylor: −2 + ½(6·0.01 + 2·0.01) = −1.960 **Off by 0.001.**

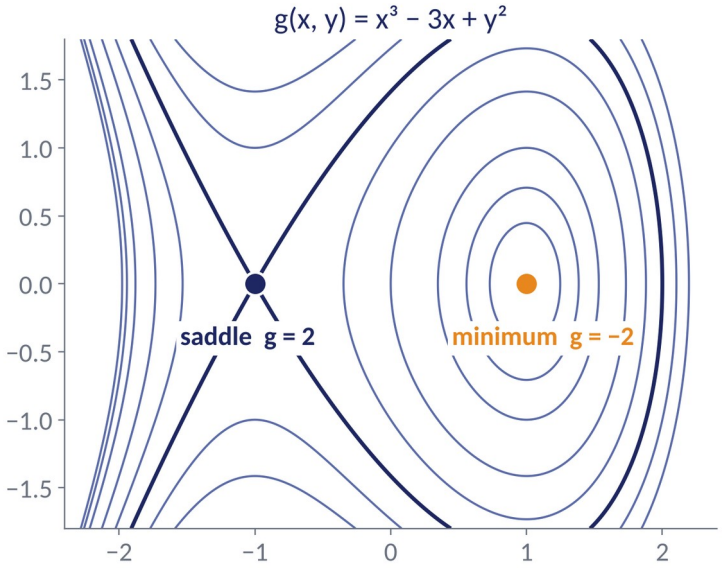
At a Critical Point, H Decides

$$\nabla f = 0, \text{ so } \quad f(x + \Delta) \approx f(x) + \tfrac{1}{2}\, \Delta^{T} H \Delta$$

H is symmetric: real eigenvalues, orthogonal eigenvectors (Module 3).

Eigenvalues of H	Type
all > 0	**minimum**
all < 0	**maximum**
mixed signs	**saddle**
some = 0	inconclusive (backup)

Each eigenvalue = the curvature along its eigenvector.



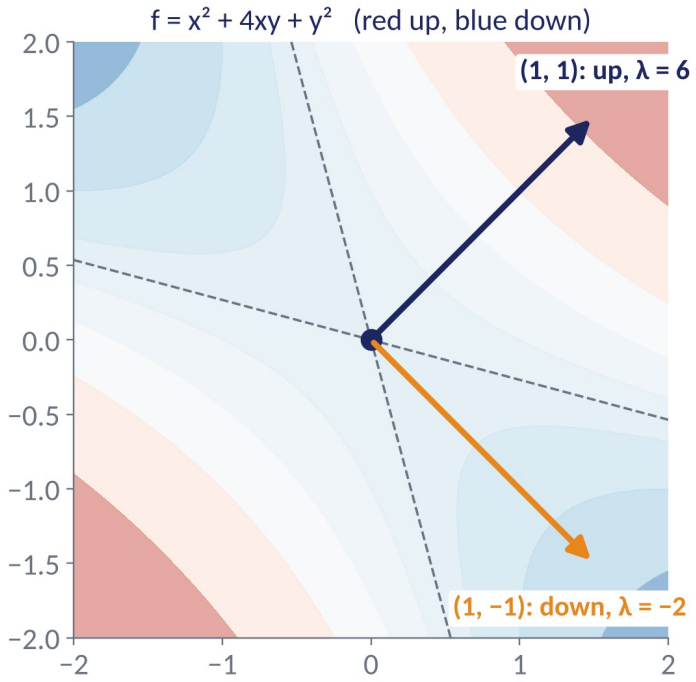
The Trap: A Positive Diagonal Isn't Enough

$$f(x, y) = x^2 + 4xy + y^2$$

$$H = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$$

Both diagonal entries positive. Minimum?

Eigenvalues 6 and −2: a saddle.
Check: f(1, −1) = −2 < 0 = f(0, 0).
The diagonal only checks east–west and north–south.



Classify the Practice Problem

$$f(x, y) = 3x^2 + 2xy + 3y^2$$

$$H = \begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}$$

Eigenvalues 8 and 4: a minimum.
Eigenvectors (1, 1) and (1, −1): the valley is steepest along the diagonal.

All eigenvalues positive everywhere: one valley, no saddles.
That's called convexity. Module 8 is built on it.

Back to the Stretched Bowl

$$f = x^2 + 3y^2, \quad H = \mathrm{diag}(2, 6)$$

λ = 2

along the x-axis

λ = 6

along the y-axis

3×

steeper in y than in x

Slide 14: the arrow hit on the axes and missed in between. The axes are the eigenvectors.

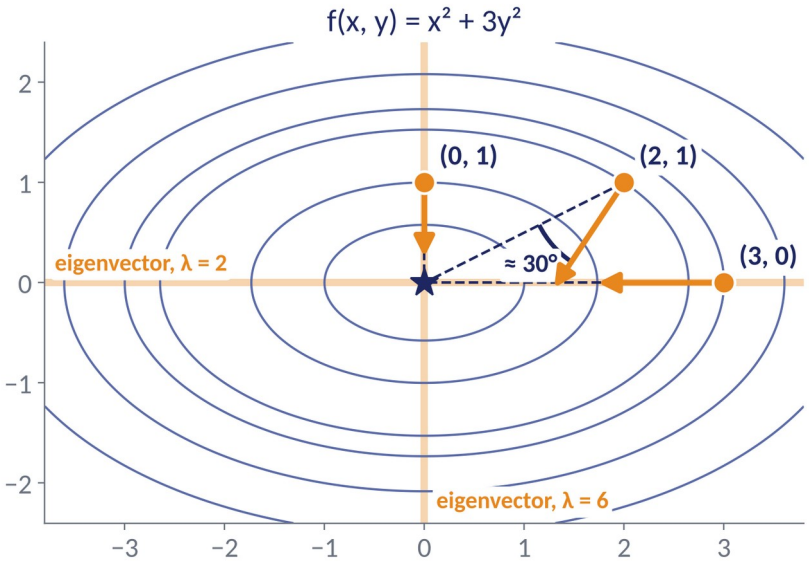
Why the Arrow Missed

$$f = \tfrac{1}{2}\, x^T H x \quad \Rightarrow \quad \nabla f = Hx$$
$$H(2,\ 1)^T = (2\cdot 2,\ 6\cdot 1) = (4,\ 6)$$

Straight to the bottom: direction −x

Steepest descent: direction −Hx

Parallel only when Hx is a multiple of x: x is an eigenvector of H.



(2, 1) → (4, 6): not a multiple, misses by 29.7°. On an axis: an eigenvector, so it hits.

How Stretched? The Condition Number

$$6 / 2 = 3$$

largest eigenvalue ÷ smallest

Round bowl: 1. The arrow always points home.

More stretched: the arrow aims worse. Module 8: this ratio sets how slowly gradient descent converges.

At work: "standardize your features" makes the bowl rounder. With millions of knobs, most critical points are saddles, not bad minima.

Next: The Gradient in One Pass

Thursday · Session 5.4: Backpropagation & Gradient Flow

How software computes the Jacobian product for 100 million knobs, and why it sometimes breaks

Questions?

Why H Is Symmetric

Mixed partials commute when the second partials are continuous (the Schwarz / Clairaut theorem):

$$\partial^2 f / \partial x \partial y = \partial^2 f / \partial y \partial x$$

Check on $f = x^2 + 4xy + y^2$: both mixed partials are 4. Optional rigor, never assessed except as bonus.

Least Squares by Setting ∇L = 0

$$L(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2 = \mathbf{x}^\top A^\top A\mathbf{x} - 2\mathbf{b}^\top A\mathbf{x} + \mathbf{b}^\top\mathbf{b}$$

$$\nabla L = 2A^\top A\mathbf{x} - 2A^\top\mathbf{b} = \mathbf{0} \implies A^\top A\,\mathbf{x} = A^\top\mathbf{b}$$

$$H = 2A^\top A \succeq 0$$

Module 4 got the normal equations with geometry; this is the calculus road. Same answer. H = $2A^TA$ is positive semi-definite, so it's always a minimum, never a saddle.

The Inconclusive Case

$$f = x^4 \quad \text{and} \quad f = -x^4$$

Both have f″(0) = 0. One is a minimum, the other a maximum.

An eigenvalue of 0 means the second-order picture is flat in that direction. You need higher-order terms to decide.

Newton's Step (Preview of Module 8)

$$\min_{\boldsymbol{\Delta}} \nabla f^\top \boldsymbol{\Delta} + \tfrac{1}{2}\boldsymbol{\Delta}^\top H \boldsymbol{\Delta} \implies \boldsymbol{\Delta} = -H^{-1}\nabla f$$

$$\text{on } f = \tfrac{1}{2}\mathbf{x}^\top H\mathbf{x}: \quad \boldsymbol{\Delta} = -H^{-1}H\mathbf{x} = -\mathbf{x}$$

Minimizing the quadratic Taylor model jumps straight to the minimum, so it fixes the 30° miss exactly. The catch: you need H, which has 10^{16} entries for 100 M parameters. Details in Module 8.